Exercise 1: Implementing the Singleton Pattern

Scenario:

You need to ensure that a logging utility class in your application has only one instance throughout the application lifecycle to ensure consistent logging.

Code:

SingletonTest.java

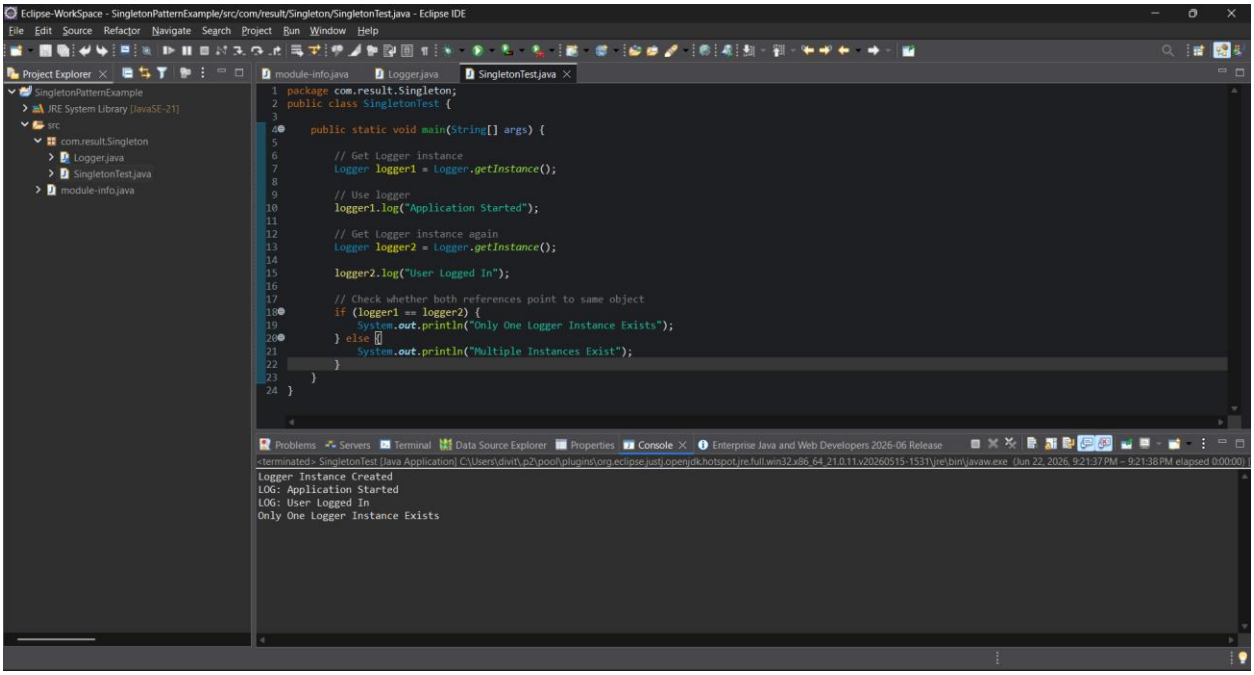
```
package com.result.Singleton;
public class SingletonTest {

    public static void main(String[] args) {

        // Get Logger instance
        Logger logger1 = Logger.getInstance();

        // Use logger
        logger1.log("Application Started");

        // Get Logger instance again
        Logger logger2 = Logger.getInstance();

        logger2.log("User Logged In");

        // Check whether both references point to same object
        if (logger1 == logger2) {
            System.out.println("Only One Logger Instance Exists");
        } else {
            System.out.println("Multiple Instances Exist");
        }
    }
}
```

Logger.java

```
package com.result.Singleton;
class Logger {

    // Step 1: Create a private static instance
    private static Logger instance;

    // Step 2: Private constructor
    private Logger() {
        System.out.println("Logger Instance Created");
    }

    // Step 3: Public method to get the instance
    public static Logger getInstance() {
        if (instance == null) {
            instance = new Logger();
        }
```

```
        return instance;
    }

    // Logging method
    public void log(String message) {
        System.out.println("LOG: " + message);
    }
}
```

Output:



Exercise 2: Implementing the Factory Method Pattern

Scenario:

You are developing a document management system that needs to create different types of documents (e.g., Word, PDF, Excel). Use the Factory Method Pattern to achieve this.

Code:

Document.java

```
package com.result.factory;

public interface Document {
    void open();
}
```

DocumentFactory.java

```
package com.result.factory;

public abstract class DocumentFactory {

    public abstract Document createDocument();}
```

ExcelDocument.java

```
package com.result.factory;

public class ExcelDocument implements Document {

    @Override
    public void open() {
        System.out.println("Opening Excel Document");
    }
}
```

ExcelFactory.java

```
package com.result.factory;

public class ExcelFactory extends DocumentFactory {

    @Override
    public Document createDocument() {
        return new ExcelDocument();
    }
}
```

PdfDocument.java

```
package com.result.factory;

public class PdfDocument implements Document {

    @Override
    public void open() {
        System.out.println("Opening PDF Document");
    }
}
```

PdfFactory.java

```
package com.result.factory;

public class PdfFactory extends DocumentFactory {

    @Override
    public Document createDocument() {
        return new PdfDocument();
    }
}
```

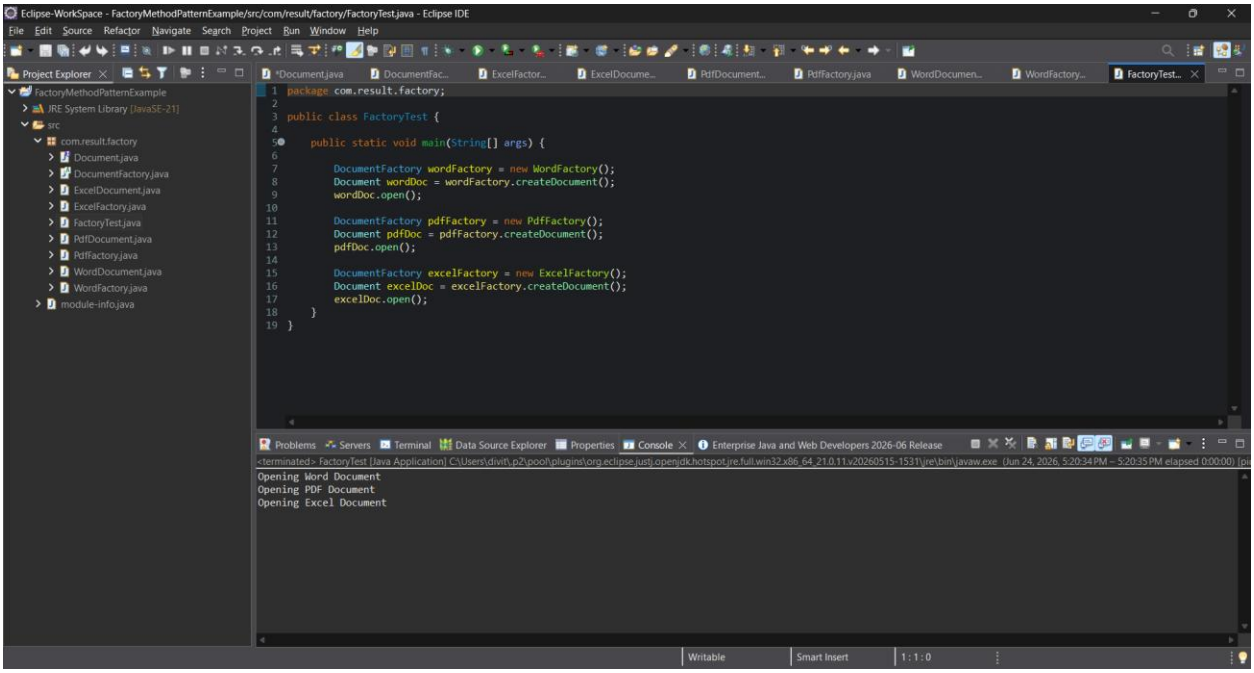
WordDocument.java

```
package com.result.factory;

public class WordDocument implements Document {

    @Override
    public void open() {
        System.out.println("Opening Word Document");
    }
}
```

WordFactory.java

```
package com.result.factory;

public class WordFactory extends DocumentFactory {

    @Override
    public Document createDocument() {
        return new WordDocument();
    }
}
```

Output:


Eclipse-WorkSpace - FactoryMethodPatternExample/src/com/result/factory/FactoryTest.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
FactoryMethodPatternExample
JRE System Library [JavaSE-21]
src
com.result.factory
Document.java
DocumentFactory.java
ExcelDocument.java
ExcelFactory.java
FactoryTest.java
PdfDocument.java
PdfFactory.java
WordDocument.java
WordFactory.java
module-info.java
*Document.java
DocumentFac...
ExcelFactor...
ExcelDocume...
PdfDocument...
PdfFactory.java
WordDocumen...
WordFactory...
FactoryTest...
1 package com.result.factory;
2
3 public class FactoryTest {
4
5 public static void main(String[] args) {
6
7 DocumentFactory wordFactory = new WordFactory();
8 Document wordDoc = wordFactory.createDocument();
9 wordDoc.open();
10
11 DocumentFactory pdfFactory = new PdfFactory();
12 Document pdfDoc = pdfFactory.createDocument();
13 pdfDoc.open();
14
15 DocumentFactory excelFactory = new ExcelFactory();
16 Document excelDoc = excelFactory.createDocument();
17 excelDoc.open();
18 }
19 }
Problems Servers Terminal Data Source Explorer Properties Console Enterprise Java and Web Developers 2026-06 Release
<terminated> FactoryTest [Java Application] C:\Users\divit\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (Jun 24, 2026, 5:20:34 PM – 5:20:35 PM elapsed 0:00:00)
Opening Word Document
Opening PDF Document
Opening Excel Document
Writable
Smart Insert
1 : 1 : 0

Exercise 3: Implementing the Builder Pattern

Scenario:

You are developing a system to create complex objects such as a Computer with multiple optional parts. Use the Builder Pattern to manage the construction process.

Code:

Computer.java

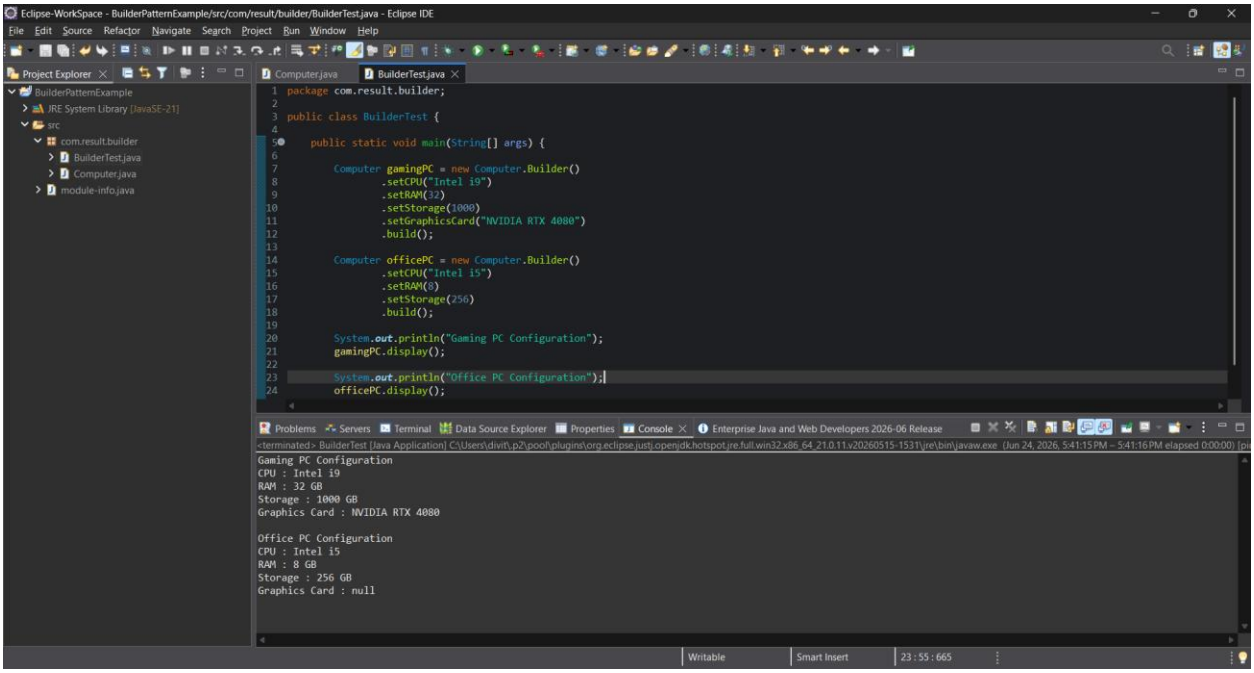
```
package com.result.builder;

public class Computer {

    private String cpu;
    private int ram;
    private int storage;
    private String graphicsCard;

    // Private Constructor
    private Computer(Builder builder) {
        this.cpu = builder.cpu;
        this.ram = builder.ram;
        this.storage = builder.storage;
        this.graphicsCard = builder.graphicsCard;
    }

    public void display() {
        System.out.println("CPU : " + cpu);
        System.out.println("RAM : " + ram + " GB");
        System.out.println("Storage : " + storage + " GB");
        System.out.println("Graphics Card : " + graphicsCard);
        System.out.println();
    }

    // Static Nested Builder Class
    public static class Builder {

        private String cpu;
        private int ram;
        private int storage;
        private String graphicsCard;

        public Builder setCPU(String cpu) {
            this.cpu = cpu;
            return this;
        }

        public Builder setRAM(int ram) {
            this.ram = ram;
            return this;
        }

        public Builder setStorage(int storage) {
```

```
            this.storage = storage;
            return this;
        }

        public Builder setGraphicsCard(String graphicsCard) {
            this.graphicsCard = graphicsCard;
            return this;
        }

        public Computer build() {
            return new Computer(this);
        }
    }
}
```

BuilderTest.java

```
package com.result.builder;

public class BuilderTest {

    public static void main(String[] args) {

        Computer gamingPC = new Computer.Builder()
                .setCPU("Intel i9")
                .setRAM(32)
                .setStorage(1000)
                .setGraphicsCard("NVIDIA RTX 4080")
                .build();

        Computer officePC = new Computer.Builder()
                .setCPU("Intel i5")
                .setRAM(8)
                .setStorage(256)
                .build();

        System.out.println("Gaming PC Configuration");
        gamingPC.display();

        System.out.println("Office PC Configuration");
        officePC.display();
    }
}
```

Output:


Eclipse-WorkSpace - BuilderPatternExample/src/com/result/builder/BuilderTest.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
BuilderPatternExample
JRE System Library [JavaSE-21]
src
com.result.builder
BuilderTest.java
Computer.java
module-info.java
Computer.java
BuilderTest.java
package com.result.builder;
public class BuilderTest {
public static void main(String[] args) {
Computer gamingPC = new Computer.Builder()
.setCPU("Intel i9")
.setRAM(32)
.setStorage(1000)
.setGraphicsCard("NVIDIA RTX 4080")
.build();
Computer officePC = new Computer.Builder()
.setCPU("Intel i5")
.setRAM(8)
.setStorage(256)
.build();
System.out.println("Gaming PC Configuration");
gamingPC.display();
System.out.println("Office PC Configuration");
officePC.display();
Problems Servers Terminal Data Source Explorer Properties Console Enterprise Java and Web Developers 2026-06 Release
<terminated> BuilderTest [Java Application] C:\Users\divit\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (Jun 24, 2026, 5:41:15 PM – 5:41:16 PM elapsed 0:00:00)
Gaming PC Configuration
CPU : Intel i9
RAM : 32 GB
Storage : 1000 GB
Graphics Card : NVIDIA RTX 4080
Office PC Configuration
CPU : Intel i5
RAM : 8 GB
Storage : 256 GB
Graphics Card : null
Writable
Smart Insert
23 : 55 : 665